

Exact Chain Elimination as Preprocessing for Transmission-Network Reduction

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Abstract—The edge-based reduction MILP carries one binary per line, so its difficulty is set by the size of the network handed to it. Much of that network carries no decision: a bus of degree two with no generator and no nearby congestion contributes a binary whose only effect is to lengthen a chain. We eliminate such buses before the MILP by a Kron reduction (Schur complement of the susceptance matrix), restricted to maximal chains so that each elimination produces exactly one equivalent series line and no fill-in. The elimination is exact – boundary angles and boundary line flows are unchanged in every scenario – and its injection transfer is a closed-form split of the eliminated load between the chain’s two anchors. Eligibility is screened so that no eliminated line is ever one whose congestion the reduction is required to preserve. After the MILP solves on the boundary network, the clustering is unfolded to a full-size assignment on the original case, so every feasibility and DC-OPF validation runs unmodified against the true network. We state what the composition preserves exactly, and the one thing it costs: a chain becomes a single binary and is therefore merged or not as a unit.

Index Terms—Transmission-network reduction, Kron reduction, Schur complement, DC power flow, mixed-integer linear programming, network equivalents.

I. SETTING

We use the notation of the companion note on the edge-based reduction model. A connected network has N buses and L lines, node–line incidence $E \in \mathbb{R}^{N \times L}$, diagonal line-susceptance matrix $D^x \in \mathbb{R}^{L \times L}$, and susceptance matrix

$$B := ED^xE^T \in \mathbb{R}^{N \times N}, \quad (1)$$

a weighted Laplacian. Over a horizon of S scenarios, scenario s has injections $p^{(s)} \in \mathbb{R}^N$, angles $\hat{\theta}^{(s)}$ and flows $\hat{f}^{(s)}$ with

$$B\hat{\theta}^{(s)} = p^{(s)}, \quad \hat{f}^{(s)} = D^xE^T\hat{\theta}^{(s)}, \quad (2)$$

gauged by $\hat{\theta}_{j_0}^{(s)} = 0$ at a reference bus j_0 . Each line l has a rating $\bar{f}_l > 0$ and an accuracy tolerance ϵ_l , and $\tau \in (0, 1]$ is the near-limit threshold: line l is *protected*, and hence forced external by the MILP, iff

$$\max_s |\hat{f}_l^{(s)}|/\bar{f}_l \geq \tau \quad (3)$$

or l is exactly at its rating in some scenario. The reduction MILP assigns a binary c_l to every line, so its size grows with L ; the purpose of this note is to shrink (N, L) before that model is built, without changing what it is allowed to conclude.

II. EXACT ELIMINATION

Partition the buses into a retained (*boundary*) set $\mathcal{R}$ and an eliminated set $\mathcal{K}$ with $|\mathcal{K}| = k$, and order (2) accordingly:

$$\begin{bmatrix} B_{\mathcal{R}\mathcal{R}} & B_{\mathcal{R}\mathcal{K}} \\ B_{\mathcal{K}\mathcal{R}} & B_{\mathcal{K}\mathcal{K}} \end{bmatrix} \begin{bmatrix} \theta_{\mathcal{R}} \\ \theta_{\mathcal{K}} \end{bmatrix} = \begin{bmatrix} p_{\mathcal{R}} \\ p_{\mathcal{K}} \end{bmatrix}. \quad (4)$$

For a connected network and $\mathcal{K} \subsetneq \{1, \dots, N\}$, the principal submatrix $B_{\mathcal{K}\mathcal{K}}$ of the Laplacian B is nonsingular (indeed positive definite), so the second block row gives $\theta_{\mathcal{K}} = B_{\mathcal{K}\mathcal{K}}^{-1}(p_{\mathcal{K}} - B_{\mathcal{K}\mathcal{R}}\theta_{\mathcal{R}})$ and substitution into the first yields

$$\tilde{B}\theta_{\mathcal{R}} = \tilde{p}, \quad \tilde{B} := B_{\mathcal{R}\mathcal{R}} - TB_{\mathcal{K}\mathcal{R}}, \quad \tilde{p} := p_{\mathcal{R}} - Tp_{\mathcal{K}}, \quad (5)$$

with the *transfer matrix*

$$T := B_{\mathcal{R}\mathcal{K}}B_{\mathcal{K}\mathcal{K}}^{-1} \in \mathbb{R}^{|\mathcal{R}| \times k}. \quad (6)$$

Two properties matter here. First, (5) is not an approximation: $\theta_{\mathcal{R}}$ solves the reduced system if and only if it is the boundary part of the solution of (4), so every line with both endpoints in $\mathcal{R}$ retains its flow exactly, in every scenario. Second, T and $\tilde{B}$ depend only on topology and susceptances, so they are computed once and applied to all S scenarios; only $\tilde{p}$ is scenario-dependent. A Schur complement of a weighted Laplacian is again a weighted Laplacian, so $\tilde{B}$ is realizable as a physical network of susceptances – but in general a dense one, which is why $\mathcal{K}$ is restricted next.

III. RESTRICTION TO CHAINS

Eliminating an arbitrary $\mathcal{K}$ fills in $\tilde{B}$ with equivalent admittances between buses that were never connected, which both destroys sparsity and creates lines the reduction model has no original counterpart for. We therefore eliminate only sets whose Schur complement is again a single series line.

Call a bus *eligible* under the rules of Section IV, and let $\mathcal{K}$ be the union of the eligible buses. Since every eligible bus has degree exactly two, the subgraph induced on $\mathcal{K}$ has maximum degree two and decomposes into disjoint simple paths and simple cycles. A maximal path

$$x - b_1 - b_2 - \dots - b_{k_c} - y, \quad x, y \notin \mathcal{K}, \quad x \neq y, \quad (7)$$

is a *chain*: x, y are its anchors and $b_1, \dots, b_{k_c}$ its interior. Write its $k_c + 1$ lines in walk order with susceptances $\beta_0, \dots, \beta_{k_c}$ and series reactances $r_i := 1/\beta_i$, and let $R_c := \sum_{i=0}^{k_c} r_i$.

Because an interior bus has degree two *in the full network*, it has no line outside its own chain. Hence $B_{\mathcal{K}\mathcal{K}}$ is block diagonal over chains, $B_{\mathcal{R}\mathcal{K}}$ couples each block only to its two anchors, and the global Schur complement (5) decouples into one independent local computation per chain. For a chain, the local Laplacian on $\{x, b_1, \dots, b_{k_c}, y\}$ is a path Laplacian, whose Schur complement onto the two anchors is

$$\tilde{B}_c = \begin{bmatrix} \beta_c & -\beta_c \\ -\beta_c & \beta_c \end{bmatrix}, \quad \beta_c = \left(\sum_{i=0}^{k_c} \frac{1}{\beta_i} \right)^{-1} = \frac{1}{R_c}, \quad (8)$$

that is, one equivalent line x – y of the classical series susceptance. The boundary network is therefore the original one with

each chain's interior buses and lines deleted and one equivalent line added, and nothing else changed:

$$\tilde{N} = N - \sum_c k_c, \quad \tilde{L} = L - \sum_c k_c, \quad (9)$$

one bus and one line removed per eliminated bus, hence one binary removed per eliminated bus. Two chains sharing an anchor, or sharing both anchors, need no special treatment: they simply contribute two equivalent lines, exactly as parallel circuits already do in the original data.

The transfer (6) likewise has a closed form on a chain. Eliminating the interior redistributes each interior injection between the two anchors in inverse proportion to its electrical distance from each,

$$\tilde{p}_x = p_x + \sum_{i=1}^{k_c} \frac{\sum_{j \geq i} r_j}{R_c} p_{b_i}, \quad \tilde{p}_y = p_y + \sum_{i=1}^{k_c} \frac{\sum_{j < i} r_j}{R_c} p_{b_i}, \quad (10)$$

the two weights summing to one, so injection is conserved. For a single interior bus between susceptances β_0, β_1 this reads $\tilde{p}_x = p_x + \beta_0 p_{b_1} / (\beta_0 + \beta_1)$: the electrically stiffer anchor absorbs the larger share. In the implementation T is obtained from the local Schur complement rather than from (10) directly, since the same factorization produces β_c and T together, but the two agree.

Two structures are excluded here rather than eliminated. A cycle lying entirely inside $\mathcal{K}$ has no anchor pair and no series equivalent. A path whose two ends are the *same* anchor would Schur-complement to a shunt at that bus, which a series-branch data model cannot express. Both are left in the boundary network untouched, where the MILP treats them as ordinary buses and lines.

IV. ELIGIBILITY

A bus b is eligible iff all of the following hold. Each rule protects one property of the construction above.

(E1) $\deg(b) = 2$. This is what makes the Schur complement a single series line (8) and produces no fill-in: the equivalent line joins two buses that were already connected through b .

(E2) *No generator at b* . An eliminated bus must carry no dispatch variable, so that the reduced DC-OPF's generator set, limits and cost data are literally unchanged. With $p = g - d$ and $g_b = 0$, the transfer (10) moves load only.

(E3) $b \neq j_0$. The reference bus pins the gauge in (2), in the MILP and in every validator; excluding it outright avoids remapping the reference into the boundary network.

(E4) *The two incident lines lead to two distinct neighbours*. A degree-two bus whose two lines run to the same neighbour is a parallel circuit terminating in a dead end, not a pass-through; eliminating it yields a shunt, not a series line, and it fails the $x \neq y$ requirement of Section III.

(E5) *Both incident lines satisfy* $\max_s |\hat{f}_l^{(s)}| / \bar{f}_l < \tau$. This is the rule that makes elimination and congestion non-interacting. Comparing with (3), (E5) says precisely that no line of any chain is ever protected. An eliminated line therefore cannot be one the reduction was obliged to keep external and reproduce, and no chain can hide a congested corridor inside an equivalent line – structurally, not as a numerical accident. The screen

is taken over the whole horizon, not over whatever scenario subset later supplies the MILP's physics constraints, so a chain is never eliminated on the strength of a favourable subset of hours.

Rules (E1)–(E4) are structural and (E5) is operational; only (E5) depends on the loading data, and only through the same worst-case criterion the MILP itself uses to decide protection. A minimum chain length may additionally be imposed: a one-bus chain removes the least size for the same per-chain coarsening cost (Section VIII), so raising the minimum trades size reduction for a finer feasible set.

V. DATA FOR THE EQUIVALENT LINE

The equivalent line inherits a rating and a tolerance from the chain it replaces:

$$\bar{f}_c = \max_{l \in c} \bar{f}_l, \quad \epsilon_c = \min_{l \in c} \epsilon_l. \quad (11)$$

The tolerance is the minimum, not the tolerance of the most loaded line and not a fresh fraction of $\bar{f}_c$ – either of which would silently widen it. The reason is the unfolding of Section VII: a chain whose equivalent line stays external is reinserted with its original lines, and each of those is then held to its *own* ϵ_l by the full-network validation. Taking the minimum guarantees no reinserted line is ever asked to fit inside a window narrower than the one its equivalent was allowed. The rating being the maximum is harmless, because the MILP's window on that line,

$$[\max\{-\bar{f}_c, \hat{f}_c - \epsilon_c\}, \min\{\bar{f}_c, \hat{f}_c + \epsilon_c\}], \quad (12)$$

is governed by the ϵ_c term whenever ϵ_c is tight, and by (E5) no chain line is near its rating in the first place.

VI. THE BOUNDARY PROBLEM

The boundary case $(\tilde{B}, \tilde{p}, \tilde{f}, \epsilon)$ is an ordinary transmission-reduction instance, and the edge-based MILP of the companion note is applied to it verbatim: binaries c_l on the $\tilde{L}$ boundary lines, lifted angles ϑ , external-flow windows, and the bounded internal transfer g^{int} . Nothing in that model is aware of the elimination. Its output is a clustering of the $\tilde{N}$ boundary buses, recovered as the connected components of $\{l : c_l = 1\}$.

VII. UNFOLDING

The boundary clustering is composed with the elimination map into a full-size assignment on the *original* N buses, by one rule per chain, applied to the chain's equivalent line x – y :

(U1) *The equivalent line ended up internal*, i.e. x and y fell in the same cluster. Then every interior bus joins that cluster. This adds no external line and no window constraint: inside a cluster all lifted angles are equal, the enclosed lines carry zero flow by construction, and the corresponding power movement is absorbed by the internal transfer g^{int} , whose role is exactly to redistribute injections within a cluster.

(U2) *The equivalent line stayed external*. Then every interior bus becomes its own cluster and the chain's original lines return with their original susceptances, ratings, injections and tolerances. Nothing was discarded in this case: the chain was

folded algebraically for the solve only, and the outcome is the same as if the elimination had never touched it.

Buses never eliminated – including those in the excluded cycles and self-anchored paths of Section III – need no rule; they are ordinary boundary buses.

The point of unfolding is what it makes possible downstream. The result is a full-size assignment on the original case, so the feasibility check, the flow-window benchmark over the whole horizon, and the reduced DC-OPF validation all run against the true original network, unmodified and unaware that any elimination took place. No accuracy claim rests on the Kron step: it changes which clusterings the MILP can reach, never the standard against which the chosen one is judged.

VIII. WHAT IT PRESERVES, AND WHAT IT COSTS

Preserved exactly: boundary-bus angles and the flows on all boundary lines, in every scenario (Section II); the total injection, redistributed in closed form and not approximated (10); the generator set and cost data (E2); the reference gauge (E3); and the protection status of every line, since no eliminated line was ever protected (E5).

Preserved in a weaker but sufficient sense: the accuracy guarantee itself. It is not inherited from the boundary solve but re-established on the full network after unfolding, which is stronger than inheriting it.

The cost is one thing, and it is structural. In the full model each of a chain's $k_c + 1$ lines carries its own binary, so the MILP may merge part of a chain – absorbing b_1 into x 's cluster while leaving b_2 outside. After elimination the whole chain is a single binary and is merged or not as a unit. The reachable set of clusterings is therefore a strict subset of the full model's, and the reduction attainable at a given tolerance ϵ is weakly worse. What is bought is a smaller mixed-integer program: $\sum_c k_c$ fewer binaries and the same number of fewer buses (9). Whether that trade pays is a property of the case and the time budget, and is to be measured by comparing against a run with the elimination disabled, not assumed.

Finally, the relation to classical equivalents. Sections II–III are a Ward-type reduction, and the usual objection to Ward equivalents applies to them in general: eliminating an arbitrary bus set invents equivalent admittances between retained buses and makes the equivalent's parameters an artifact of the operating point used to build it. Neither happens here. The eliminations are confined to chains, so the only admittance created is the series equivalent of a path that already connected its two anchors; and the transfer (10) is exact for every scenario simultaneously, not fitted to one. The reduction MILP downstream, in turn, creates no equivalent admittances at all – retained lines keep their original susceptances.